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年級: 微 91

科目: 概率論與隨機過程

1. (a) X = the number of modems in use at the given time

$$P_X(k) = \begin{cases} \binom{1000}{k} 0.01^k \cdot 0.99^{1000-k}, & k=0, 1, 2, \dots, 49 \\ 1 - \sum_{k=0}^{49} \binom{1000}{k} 0.01^k \cdot 0.99^{1000-k}, & k=50 \\ 0, & \text{otherwise} \end{cases}$$

(b) Y = the number of customers that need a connection

$$P_Y(k) = \binom{1000}{k} 0.01^k \cdot 0.99^{1000-k} \approx e^{-\lambda} \frac{\lambda^k}{k!}$$

$$\text{Let } \lambda = 1000 \times 0.01 = 10, \Rightarrow P_Y(k) = \frac{10^k \cdot e^{-10}}{k!}$$

(c) $P(\text{more customers needing a connection than there are modems})$

$$= 1 - \sum_{k=0}^{50} \binom{1000}{k} 0.01^k \cdot 0.99^{1000-k} \approx 1 - \sum_{k=0}^{50} \frac{10^k \cdot e^{-10}}{k!}$$

$$2. (a) \text{ As } \sum_x P_X(x) = 1, \quad 1 = \sum_x P_X(x) = \sum_{k=-3}^3 \frac{k^2}{a} = \frac{28}{a} \Rightarrow a = 28$$

$$E[X] = \sum_{k=-3}^3 \frac{k^3}{a} = 0$$

$$(b) Z = (X - E[X])^2 = X^2$$

$$P_Z(x) = \begin{cases} \frac{2x}{a}, & x=1, 4, 9 \\ 0, & \text{otherwise} \end{cases} = \begin{cases} \frac{x}{14}, & x=1, 4, 9 \\ 0, & \text{otherwise} \end{cases}$$

$$(c) \text{ variance of } X = \text{var}(X) = E[Z] = \sum_x x P_Z(x) = \frac{1^2}{14} + \frac{4^2}{14} + \frac{9^2}{14} = 7$$

$$(d) \text{ var}(X) = \sum_x (x - E[X])^2 P_X(x) = \sum_x x^2 P_X(x) = \sum_{k=-3}^3 \frac{k^4}{28} = \sum_{k=1}^3 \frac{k^4}{14} = 7$$

3. X = the number of games that are played
 Obviously, if $P_X(k) > 0$, then $3 \leq k \leq 5$

$$\begin{aligned} & \cancel{P_X(3) = p^3, P_X(4) = p \cdot \binom{3}{1} (1-p)p^2 = 3(1-p)p^3, P_X(5) =} \\ & \begin{cases} P_X(3) = p^3 + (1-p)^3 = 1 - 3p + 3p^2 \\ P_X(4) = p \cdot \binom{3}{1} (1-p)p^2 + (1-p) \cdot \binom{3}{1} p(1-p)^2 = 3p(1-p)(2p^2 - 2p + 1) \\ P_X(5) = p \binom{4}{2} (1-p)^2 p^2 + (1-p) \binom{4}{2} p^2 (1-p)^2 = 6(1-p)^2 p^2 \\ P_X(k) = 0, k \notin \{3, 4, 5\} \end{cases} \end{aligned}$$

$$\begin{aligned} E[X] &= 3P_X(3) + 4P_X(4) + 5P_X(5) = 4(P_X(3) + P_X(4) + P_X(5)) + P_X(5) - P_X(3) \\ &= 4 + 6(1-p)^2 p^2 - 1 + 3p - 3p^2 \\ &= 3 + 6(1-p)^2 p^2 + 3(1-p)p \\ &\leq 3 + 6 \cdot \left(\frac{1-p+p}{2}\right)^4 + 3 \cdot \left(\frac{1-p+p}{2}\right)^2 = 3 + \frac{3}{8} + \frac{3}{4} = 4\frac{1}{8} = \frac{33}{8} \end{aligned}$$

(Equality occurs if and only if $1-p=p \Leftrightarrow p=\frac{1}{2}$)

4. Obviously, $P_X(x) = \begin{cases} p, & x=1 \\ 1-p, & x=-1 \\ 0, & \text{otherwise} \end{cases} \Rightarrow E[C^X] = cp + c^{-1}(1-p) = 1$

~~$E[C^X] = cP_X(1) + c^{-1}P_X(-1)$~~

$$\Leftrightarrow pc^2 - c + (1-p) = 0$$

$$\Leftrightarrow c = \frac{1 \pm \sqrt{1-4p(1-p)}}{2p} = \frac{1 \pm |2p-1|}{2p} = \frac{1 \pm (2p-1)}{2p}$$

$$\Rightarrow c = 1 \text{ or } c = \frac{1-p}{p} = \frac{1}{p} - 1$$

$$\text{As } c \neq 1, c = \frac{1}{p} - 1$$